

The Use of Motion Detector Camera in Uveal Melanoma Stereotactic Radiotherapy

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Abstract— The standard stereotactic radiotherapy treatment for uveal melanoma uses several 6MV photon beams collimated through cones that focus the beam to the lesion. The patient head is immobilized using the head rest or a mask to minimize head motion during the treatment. Furthermore, the patient is also asked to stare at an object of regard (OOR) during treatment to focus the beams to the lesion and not irradiate normal parts in the orbit, e.g., the optic nerve. This setup usually encounters difficulties if the patient is weak and gets drowsy during the treatment or if the case involves a pediatric patient. The De La Salle University Medical Physics Department is developing a tool that will ensure accurate treatment during this procedure and to minimize dose to other normal tissues. A motion detector camera placed at the object of regard (OOR) is interfaced with a computer that sets a limit on the deviation of the eyeball from a preset position. Once the patient moves the eyeball beyond this limit, the program sends a signal to the linear accelerator to stop the treatment. A computer program will be developed to interface the detector camera to the linear accelerator.

Keywords— Motion detector camera, uveal melanoma, computer interface

I. INTRODUCTION

Uveal melanoma is a rare tumor originating in melanocytes of the choroid, ciliary body, or iris and can result in visual deficit and metastatic death. If left untreated, death typically occurs within months after metastases are detected clinically.

Uveal melanoma is more common in lightly pigmented persons than in nonwhite races[1]. The incidence of the disease among African Americans is less than one eighth of the incidence in whites.

Enucleation was for a long time the only treatment option possible. The procedure would sacrifice vision in order to save life of the patient. About half a century ago, the radiation was first introduced as a noninvasive treatment option to avoid enucleation. However, some side effects were observed, including vision loss and facial skin burns. Many of the first radiation therapy patients relapsed and manifested metastasis a few months after the procedure. This old procedure involved, at best, small fields directed towards the back of the orbit.

Stereotactic radiosurgery/radiotherapy (SRS/SRT) uses multiple high energy beams converging to the tumor with extreme precision[2]. The treatment can be done with a linear accelerator or with the Gamma Knife. Virtually no side effects have been recorded and the rate of relapse and metastasis have dramatically reduced.

SRS/SRT differs from conventional radiotherapy, which involves exposing large areas of intracranial tissue to relatively broad fields of radiation over a number of sessions. Highly focused convergent beams are delivered in a single session so that only the desired target is radiated, sparing adjacent structures. This setup is repeated over several days until the desired target dose is achieved.

II. CONSTRUCTION OF THE CAMERA

The setup for SRT/SRS involves a head immobilizer and a blinking light that the patient stares at during the treatment. This setup usually encounters difficulties if the patient is weak and gets drowsy during the treatment or if the case involves a pediatric patient. The De La Salle University Medical Physics Department is developing a tool that will ensure accurate treatment during this procedure and to minimize dose to other normal tissues. A motion detector camera placed at the object of regard (OOR), or the blinking light is interfaced with a computer that sets a limit on the deviation of the eyeball from a preset position. Once the patient moves the eyeball beyond this limit, the program sends a signal to the linear accelerator to stop the treatment. A computer program will be developed to interface the detector camera to the linear accelerator.

This research is still currently being conducted. We are currently looking at small video cameras sold in the market that have motion detection features. We will explore the use of LABView and C++ for the computer interface. The computer program will be put in a software that will be included in the linear accelerator interlock. Testing of the procedure will be conducted at the St. Lukes Medical Center in Quezon City, Manila. During the testing, the computer will be attached to the motion detector camera. A motion limit parameter will be set on an object that the camera will detect. Then the object will be moved beyond

the set limit to see if the Linear accelerator interlock will work to turn off the machine.

III. CONCLUSIONS

Stereotactic radiotherapy/radiosurgery has indeed proven to be very accurate for many brain lesions. However, the versatility of the procedure is stretched to its limit when treatment for uveal melanoma is carried out – owing to the fact that the eyeball might move a lot during the treatment. The introduction of a motion detector camera that is attached to the machine interlock might solve this problem. This gadget will further increase the accuracy of SRS and will minimize possible radiation to critical structures such as the lens and the optic nerve.

REFERENCES

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